

From Ground Station to Flying 5G Relay

Progress on a Drone-Oriented 5G Testbed

Research Progress Recap

Presenter: Research Team

Slide 1 — The idea in one picture

Project Concept

Instead of carrying an entire heavy 5G base station on a drone, we **separate the lightweight radio** parts from the heavy computing intelligence on the ground.

The core network brain stays safe on the ground, while the drone serves as a lightweight flying relay.

```
graph TD
    subgraph Ground["Ground Station"]
        A["Core Network + CU<br>(Heavy Computing)"]
    end
    subgraph Air["Drone Relay"]
        B["DU + USRP B210 Radio<br>(Lightweight RF Payload)"]
    end
    subgraph User["Emergency Area"]
        C["Nothing Phone User<br>(PWS Warning Alert)"]
    end
    A -->|"WireGuard (F1-C/U)"| B
    B -.->|"5G n78 Access Link"| C
    style Ground fill:#1e293b,stroke:#3b82f6,stroke-width:2px,color:#f8fafc
    style Air fill:#1e293b,stroke:#10b981,stroke-width:2px,color:#f8fafc
    style User fill:#1e293b,stroke:#f43f5e,stroke-width:2px,color:#f8fafc
```

Slide 2 — Why this project matters

Disaster & Emergency Coverage

Temporary coverage is vital after **floods, forest fires, or critical outages** where ground infrastructure is destroyed.

A drone can **restore emergency coverage within minutes**. The challenge is keeping airborne equipment light and power-efficient.

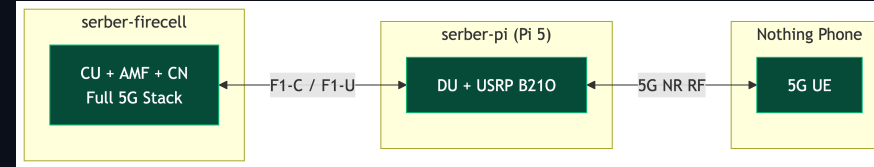
```
flowchart TD
    Disaster["Disaster Event<br>(Floods, Fires, Outages)"] --> Outage["Ground Towers Broken<br>(No Cellular Signal)"]
    Outage --> Rescue["Drone 5G Relay<br>(Deploys in Minutes)"]
    Rescue --> Recovery["Emergency Coverage Restored<br>(PWS Warning Sent)"]
    style Disaster fill:#1e293b,stroke:#ef4444,stroke-width:2px,color:#f8fafc
    style Outage fill:#1e293b,stroke:#ef4444,stroke-width:2px,color:#f8fafc
    style Rescue fill:#1e293b,stroke:#10b981,stroke-width:2px,color:#f8fafc
    style Recovery fill:#1e293b,stroke:#10b981,stroke-width:2px,color:#f8fafc
```

Slide 3 — What CU/DU split means

Architecture Concept

We divide a 5G base station into two parts:

- **Core Network & CU:** The ground-based "brain" managing user sessions, security, and protocols.
- **DU & Radio antenna:** The drone-carried "relay" transmitting actual high-speed radio waves.
- **Backhaul Link (F1):** Wireless connection linking them.

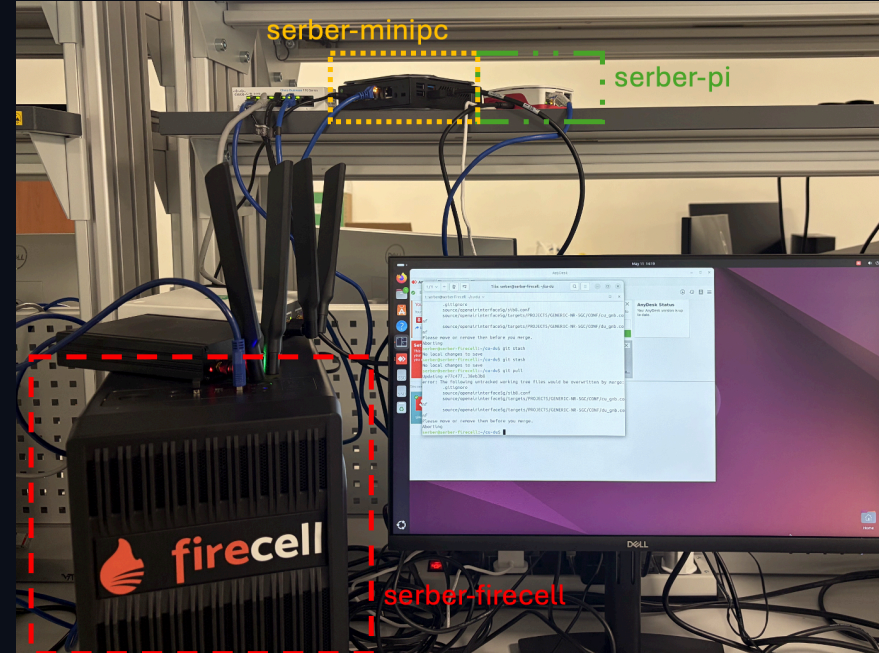


Slide 4 — The physical testbed

Hardware Setup

We validated a real-world benchtop setup using accessible hardware:

- **serber-firecell**: Ground Core & CU.
- **serber-minipc**: Remote DU unit.
- **Raspberry Pi 5**: Portable DU candidate.
- **USRP B210**: Software-programmable antenna.
- **Quectel Modem**: Wireless backhaul link.



Slide 5 — Progress timeline

Milestones Timeline

The system progressed systematically from software simulations to real-world wireless hardware loops.

Key achievements include connecting a **commercial phone**, broadcasting **emergency warnings (PWS)**, and implementing **5G WireGuard backhauling**.

```
flowchart LR
    A["Simulation"] --> B["Real Radio"] --> C["Nothing Phone"] --> D["Emergency PWS"] --> E["Ethernet Split"] --> F["Pi 5 DU"] --> G["WiFi GRE"] --> H["Quectel 5G"] --> I["Target Setup"]
    style A fill:#1e293b,stroke:#3b82f6,color:#f8fafc
    style B fill:#1e293b,stroke:#3b82f6,color:#f8fafc
    style C fill:#1e293b,stroke:#10b981,color:#f8fafc
    style D fill:#1e293b,stroke:#10b981,color:#f8fafc
    style E fill:#1e293b,stroke:#10b981,color:#f8fafc
    style F fill:#1e293b,stroke:#10b981,color:#f8fafc
    style G fill:#1e293b,stroke:#f59e0b,color:#f8fafc
    style H fill:#1e293b,stroke:#f59e0b,color:#f8fafc
    style I fill:#1e293b,stroke:#f59e0b,color:#f8fafc
```

Slide 6 — The main engineering pivots

Hurdles & Adaptations

Faced with real-world hardware limits, the project shifted direction to find viable solutions:

We resolved **protocol blocks**, **hardware overloading**, and **signal saturation** to reach a stable prototype.

1. Tegra Kernel Roadblock

Jetson Orin Nano lacked SCTP protocol support. Pivoted to x86 and Pi 5 for native kernel support.

2. Single-Radio Overload

One radio could not serve both access and backhaul. Dedicated B210 to phone, and Quectel modem to backhaul.

3. Throughput Bottleneck

Upgrading DU strength did not remove split-mode speed limits. Shifted focus to OAI scheduler tuning.

Slide 7 — What works today

Current Status

Our current system has successfully validated all key features of the split 5G architecture.

Performance optimization and final user-traffic validation over the wireless backhaul link are currently under active testing.

- **Commercial phone attached**
- **Public Warning System SIB8**
- **CU/DU split F1 separation**
- **Wireless F1 tunnel over WiFi/5G**
- **Raspberry Pi 5 DU timing fix**
- **End-to-end backhaul speed**
- **Split scheduler bottleneck fix**

Done

Done

Done

Done

Done

Testing

Testing

Slide 8 — Current target architecture

System Layout

Our unified architecture runs on a single central server:

- **serber-firecell (Ground):** Runs 5G Core, CU, and donor base station.
- **serber-minipc (Remote):** Runs DU, access B210, and Quectel modem.
- **WireGuard Tunnel:** Over the Quectel 5G link, carrying F1-C & F1-U.

```
graph TD
    subgraph Ground["Ground Station (serber-firecell)"]
        Core["OAI 5G Core"]
        CU["OAI Central Unit (CU)"]
        Donor["OAI Donor Cell (gNB)<br>(PCI 1, TAC 2)"]
        Core --- CU
        Core --- Donor
    end
    subgraph Remote["Remote Unit (serber-minipc)"]
        Quectel["Quectel 5G Modem<br>(IP 10.0.0.6)"]
        DU["OAI Distributed Unit (DU)<br>(PCI 0, TAC 1)"]
        B210["USRP B210 Access Radio"]
        DU --- B210
    end
    subgraph UE["User Device"]
        Phone["Nothing Phone"]
    end

    Donor -.->|"5G RF Link"| Quectel
    Quectel == "WireGuard F1 Link" ==> CU
    B210 -.->|"5G RF Access"| Phone

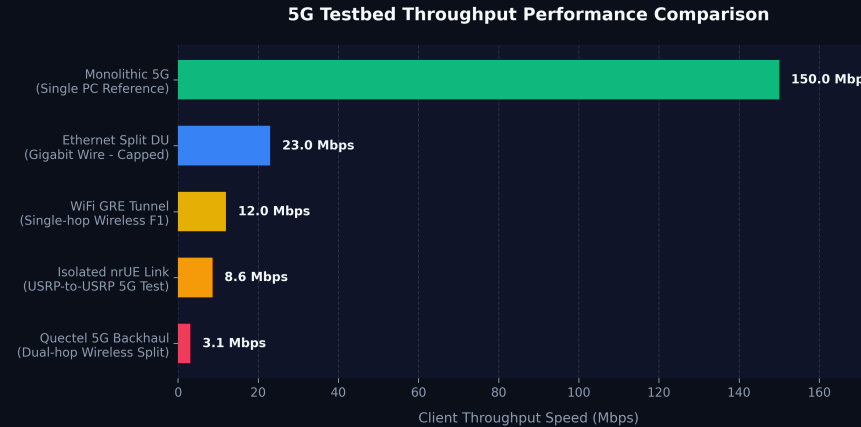
    style Ground fill:#1e293b,stroke:#3b82f6,stroke-width:2px,color:#f8fafc
    style Remote fill:#1e293b,stroke:#10b981,stroke-width:2px,color:#f8fafc
    style UE fill:#1e293b,stroke:#f43f5e,stroke-width:2px,color:#f8fafc
```

Slide 9 — Results and remaining bottleneck

Throughput Benchmarks

While monolithic 5G achieves 150 Mbps, split mode drops to **23 Mbps**. Moving the DU to stronger hardware did not increase speed.

The scheduler pins transmission rates to the lowest setting (**MCS 0**) due to delayed channel reports.



Slide 10 — What is next

Roadmap to Flight

We have moved from simulation to a real, separated 5G network with wireless backhaul.

Our next challenge is making this setup fast, reproducible, and ready for drone flights.

1. **Prove Backhaul Path:** Confirm phone traffic runs strictly over Quectel counters.
2. **Tune Scheduler:** Break the 23 Mbps limit.
3. **Package Code:** Secure configurations.
4. **Validate Pi 5 DU:** Re-test Pi DU on backhaul.
5. **Airborne Flight:** Transition to flight.